Student number:

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EEE3094S 2023: Class Test 1

- The test is open book
- Answer on the test sheet. If you run out of space, you may attach additional pages, but please note that you have done so (e.g. write "see attached pages" by the question) and make sure they are marked with your student number.
- There are four questions totalling 50 marks.
- You have one hour to complete the test.
- The last page is an information sheet.
- The information sheet includes a photo of my cat, Mouse. You are invited to draw a picture of her in the space below if you finish/ give up early and are bored. Doing so will not officially get you extra marks, but it will probably make me and the tutor marking your test happy, which might make a difference in your favour. **Also, the "best" Mouse drawing will gain the artist an unspecified reward.**

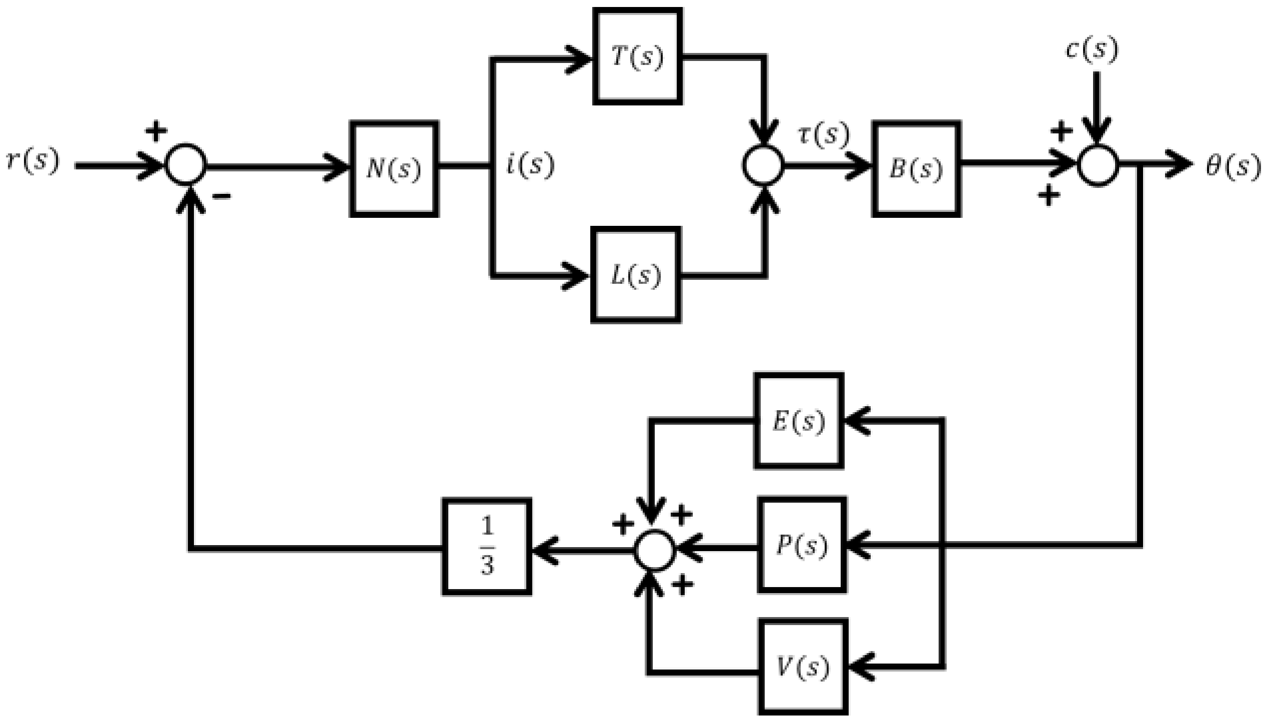
Class Test 1

Question 1 [12]

Neuromechanics is an area of biological research that studies how the nervous system controls the musculoskeletal system to bring about movement. A neuromechanics problem that the African Robotics Unit is interested in investigating is the role of the cheetah's tail in balancing its body. Gavin and I recently travelled up to the Ann van Dyk Cheetah Centre to film cheetahs being transported by bakkie. As the car bumps over the rough dirt roads around the sanctuary, the animals often lose their footing and are seen swinging their tails to recover.

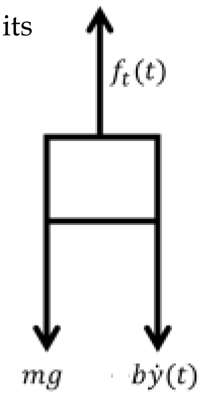
Your task is to draw a block diagram modelling the cheetah in the vehicle. Here is the information your diagram must capture:

- The cheetah can be broken down into the following subsystems: the nervous system $N(s)$, the legs $L(s)$, the tail $T(s)$, the body $B(s)$, the eyes $E(s)$, the vestibular system $V(s)$ and the proprioceptive system $P(s)$.
- The cheetah is trying to control its body angle $\theta(s)$. It uses a combination of the eyes, vestibular system and proprioceptive system to detect this angle. You may consider the sensed angle to be the *average* of these three readings.
- The nervous system considers the difference between the desired angle $r(s)$ and the sensed position, and sends a nervous impulse $i(t)$ to the legs and tail to command them to respond.
- The legs and tail respond to the impulse, producing a combined torque $\tau(s)$ that acts to correct the body angle.
- The motion of the vehicle $c(s)$ also affects the angle of the body.



Question 2 [12]

The diagram on the right illustrates the forces acting on a quadcopter. You wish to control its vertical velocity $\dot{y}(t)$.



The rotor thrust $f_t(t)$ is the input to the system. Assuming there is no wind, the aerodynamic drag will be roughly proportional to the velocity of the quadcopter (you can assign it a coefficient b). Gravity acts as a constant external disturbance. The mass of the quadcopter is 1 kg. **You can use $g = 9.8$**

a) Write a differential equation modelling the quadcopter and hence, show that the relationship between the velocity and rotor thrust can be represented by a first-order transfer function. [2]

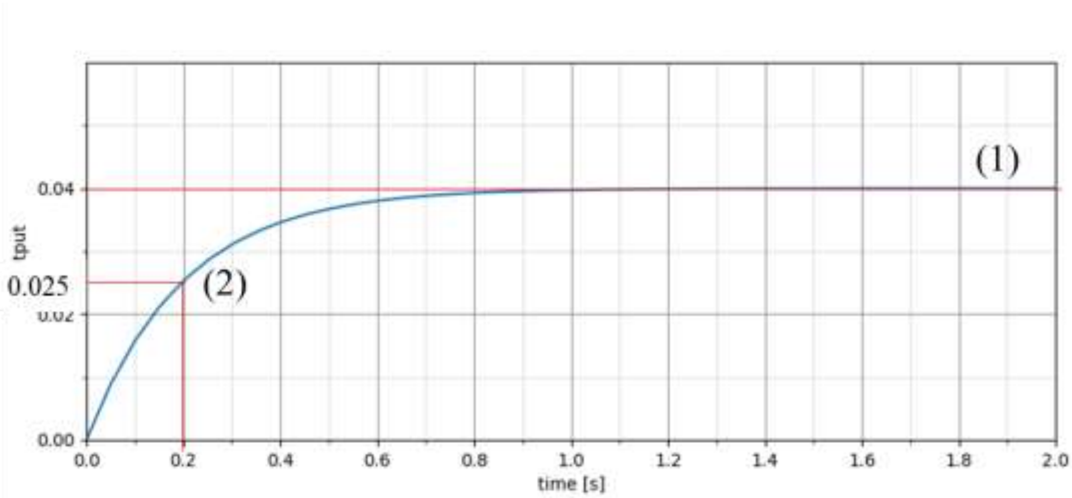
Solution:
Differential equation: $m\ddot{y}(t) = f_t(t) - b\dot{y}(t) - mg$
Even though there is acceleration in this equation, this is only the first derivative of the variable we are interested in (the velocity) so this is effectively a first-order differential equation.

Taking the Laplace transform of both sides (assuming initial rest) gives:

$$ms\dot{Y}(s) = F_t(s) - b\dot{Y}(s) - \frac{mg}{s}$$

This leads to the transfer function $G(s) = \frac{1}{ms-b}$ relating the velocity to the thrust.

b) You apply a 10 N rotor force to the resting quadcopter, resulting in the change in velocity shown below. Use this to calculate the damping coefficient b and complete the transfer function model of the system. Use *two* features of the given response to confirm your transfer function model of the system. You may annotate the plot to indicate the features of interest. [4]



Solution:
Subtracting the weight of the quadcopter (9.8) gives an input step of magnitude 0.2 N.
Considering the final value of $y(\infty) = 0.04$ (shown at point 1 on the plot), the gain of the transfer function is therefore $A = \frac{0.04}{0.2} = 0.2$
The step response reaches $0.63y(\infty) = 0.025$ at $t = 0.2$ seconds (point 2 on the plot), so $\tau = 0.2$.
This gives the transfer function $G(s) = \frac{0.2}{0.2s+1} = \frac{1}{s+5}$

c) You decide to use proportional feedback control to regulate the velocity of the quadcopter. You may assume perfect feedback. Determine the gain of the controller required to reduce the effect of gravity to a magnitude of less than 1 m/s on the output velocity. [3]

Solution:

Using the final value theorem, the effect of gravity is given by $\lim_{s\to 0} s\frac{9.8}{s}G(s)$.

This equates to $\frac{9.8}{1+K_p}$, where $K_p = \lim_{s\to 0} KG(s)$ is the position error constant (K is the controller gain).

To get $\frac{9.8}{1+K_p} \leq 1$, we need $K_p \geq 8.8$

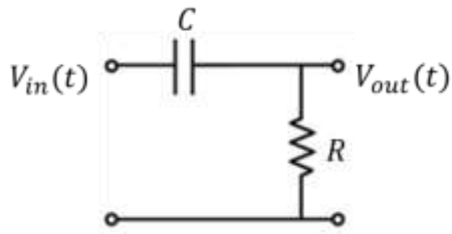
Because the gain of the system is 0.04, $0.04K = 8.8 \therefore K = 220$

d) Would it be possible to reduce the effect of gravity to zero using this type of controller? Answer in terms of the type number of the system, and suggest an alternative control approach if not. [3]

Solution:

No, gravity is interpreted as a step disturbance, meaning that the system would have to be Type 1 or greater to nullify the effect. There are no integrators (poles at zero) in the transfer function, so it is Type 0. An integral controller of the form $C(s) = \frac{K}{s}$ would raise the type number of the system and enable it to reject the disturbance completely.

Question 3 [14]



The output voltage of the circuit to the right is related to the input voltage by the differential equation

$$V_{out}(t) + RC\frac{d}{dt}V_{out}(t) = RC\frac{d}{dt}V_{in}(t)$$

and hence, the transfer function

$$G(s) = \frac{RCs}{1 + RCs}$$

You may use $\boldsymbol{R = 1\ \mathrm{k}\Omega, C = 0.1\ F}$

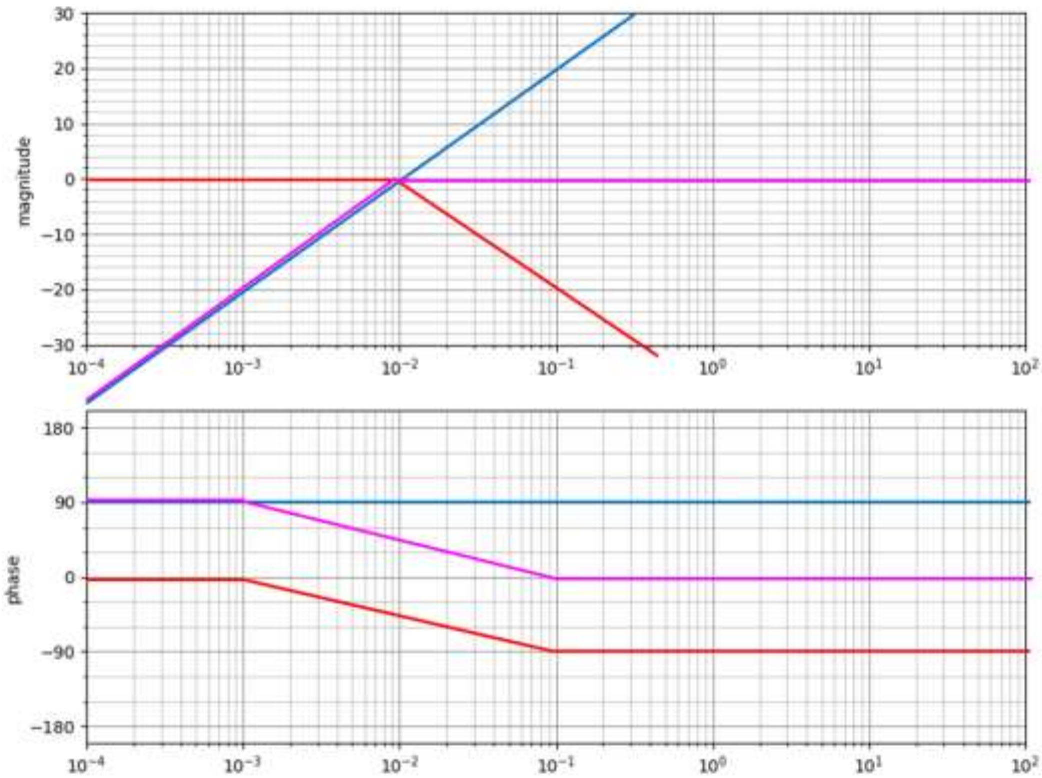
a) Roughly sketch the frequency response of this circuit as a Bode plot on the graph below. What type of filter is this? [8]

<u>Solution:</u>
I would suggest sketching the zero and pole responses separately and then combining them. Here, I will show the zero in blue, the pole in red and the combined response in pink.

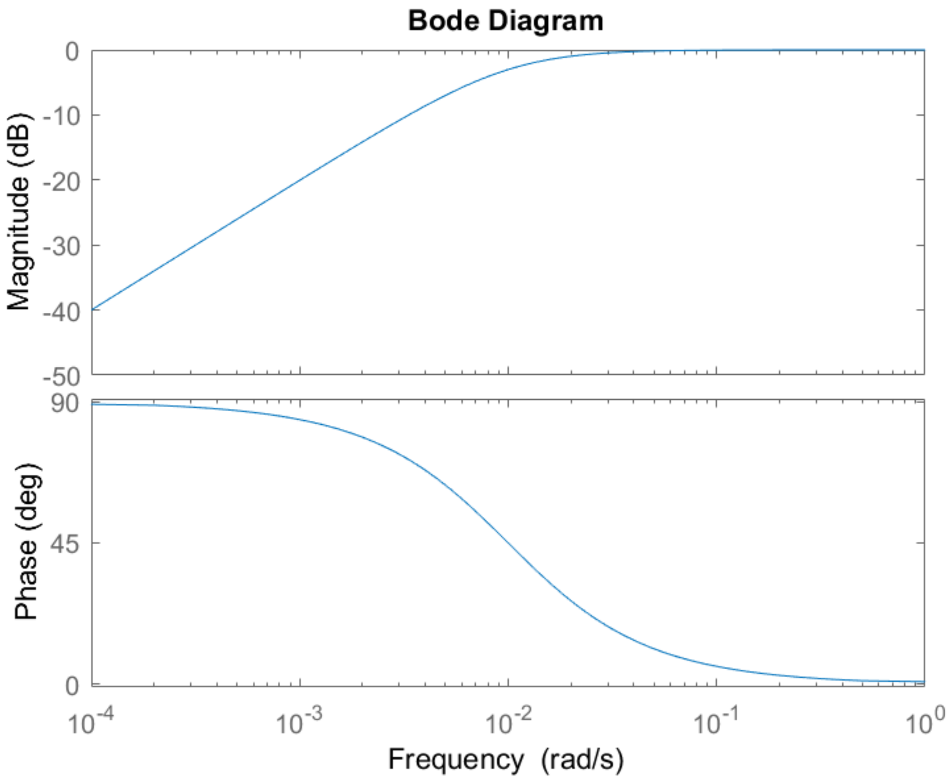
The zero is at 0 so its corner frequency is $\omega = 0$. You can't show this value on a log scale, so the lowest frequency we have is 10^{-4}. The magnitude of the zero term there is $100(10^{-4}) = 0.01$, which is $20\log_{10}(0.01) = -40$ dB. It will roll up from there at 20 dB per decade. The phase of the zero is a constant 90°

The pole is at $-\frac{1}{100} = -0.01$, so its corner frequency is $\omega = 0.01$. The magnitude of the pole term at low frequencies is close to 1 (or 0 dB), and it will then roll off at -20 dB per decade from the corner frequency. The phase of the pole starts at 0 and transitions to -90 °, hitting $-45°$ at the corner frequency.

Asymptotic drawing:



Confirmation via MATLAB:



This is a high pass filter.

b) Calculate the initial value of the unit step response of this circuit, assuming the capacitor is initially fully discharged (i.e. initial rest). [2]

Solution:

Using the initial value theorem, $V_o(0) = \lim_{s \to \infty} \frac{100s}{1+100s} = 1$

c) If the capacitance of the circuit is increased, how will this affect

i. the frequency response of the circuit? [2]

ii. the time response of the circuit? [2]

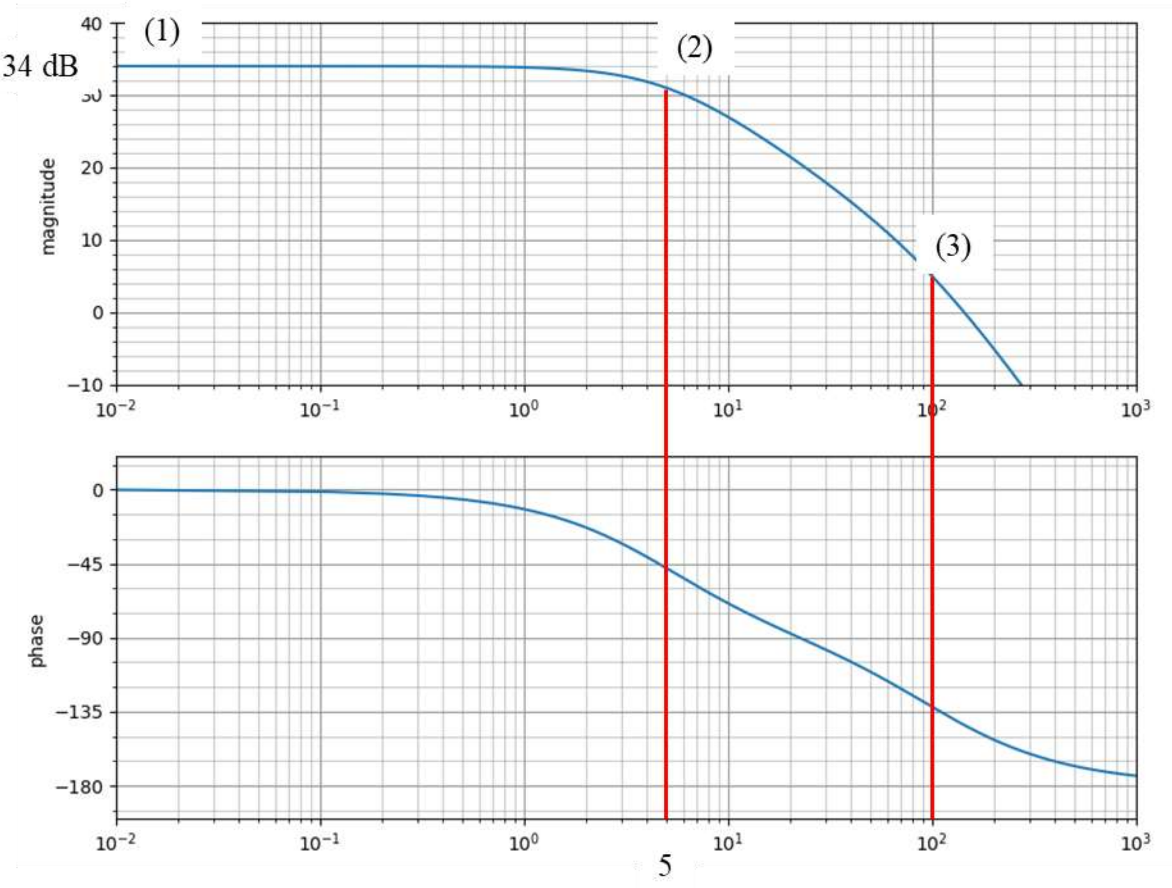
Solution:

i. Increasing the capacitance will not affect the corner frequency of the zero, but it will cause the corner frequency of the pole to decrease. This will widen the pass band of the filter to start at a lower frequency.

ii. Decreasing the corner frequency of the pole is the same as making the response 'slower' in the time domain. (E.g. the step response will have a longer rise time)

Question 4 [12]

The Bode plot below gives the frequency response of a second-order system.



a) The transfer function $G(s) = \frac{50000}{s^2+105s+500}$ is proposed as a model for the system. Confirm this model with reference to the given frequency response. You may indicate relevant features by annotating the plot. [4]

Solution:

Correction: the transfer function should be $\boldsymbol{G(s) = \frac{25000}{s^2+105s+500}}$

The gain of this transfer function is $\lim_{s \to 0} \frac{25000}{s^2+105s+500} = 50$, or $20 \log_{10}(50) = 34\ dB$. This matches the low-frequency gain shown on the plot at point 1.

The response is consistent with a system having two real poles, as two distinct corner frequencies are visible, and the magnitude has one section that rolls off at -20 db/ decade, followed by one rolling off at -40 dB/ decade.

Based on the locations of the corners between these asymptotes, and the places where the phase plot crosses $-45°$ and $-135°$ (points 2 and 3), the poles appear to be at -5 and -100. This is consistent with the denominator of $(s + 5)(s + 100) = s^2 + 105s + 500$.

b) Do you expect that the step response of the system will overshoot? Explain. [2]

Solution:

No. The system has two real poles, so it is overdamped. The response will therefore not oscillate or overshoot.

c) Is there a simplified model that will give a good approximation of the behaviour of this system? If so, suggest such a model. [2]

Solution:

Yes. The second pole at -100 is much more than five times faster than the dominant pole at -5, so we can replace the factor $(s + 100)$ in the denominator with a gain of 100 to give the first-order model:

$$G_r(s) = \frac{25000}{100(s+5)} = \frac{250}{s+5}$$

d) The system is now placed in negative feedback with a proportional controller. You may assume perfect feedback.

i. Calculate the position error constant, and hence, the steady-state error with respect to a unit step if the controller has a gain of 1. [2]

ii. Calculate the controller gain at which the closed loop system is critically damped. [2]

Solution:

i. The position error constant is given by $\lim_{s\to 0} G(s) = 50$. Consequently, the steady-state error for a unit step input will be $\frac{1}{1+50} = 0.196$

ii. The characteristic equation of the closed loop system is $s^2 + 105s + 500 + K = 0$

Using the quadratic equation, the poles of this system are at $\frac{-105 \pm \sqrt{105^2 - 4(500+K)}}{2}$

The system will be critically damped when the value under the square root is zero, resulting in two real poles at the same location. Therefore, when $105^2 - 4(500 + K) = 0$

We can solve this to give $K = 2000 - 105^2 = -9025$, so this is only possible with a negative gain.

Information

$$\frac{A\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$\omega_d = \omega_n\sqrt{1-\zeta^2}$$

$$\cos\theta = \zeta$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y_p = A\left(1 + e^{-\zeta\pi/\sqrt{1-\zeta^2}}\right)$$

$$t_r = \frac{\pi - \theta}{\omega_d}$$

$$t_{k\%} = \frac{\ln\left(\frac{k}{100}\sqrt{1-\zeta^2}\right)}{-\zeta\omega_n}$$

Laplace transforms of signals:

$y(t)$	$Y(s)$	Comments
$\sigma(t)$	$\frac{1}{s}$	unit step
$t\sigma(t)$	$\frac{1}{s^2}$	unit ramp
$e^{-at}\sigma(t)$	$\frac{1}{(s+a)}$	decaying exponential if $a > 0$
$t^n e^{-at}\sigma(t)$	$\frac{n!}{(s+a)^{(n+1)}}$	n^{th} order exponential
$\sin(\omega t)\sigma(t)$	$\frac{\omega}{s^2+\omega^2}$	sinusoid
$\cos(\omega t)\sigma(t)$	$\frac{s}{s^2+\omega^2}$	cosinusoid
$e^{-at}\sin(\omega t)\sigma(t)$	$\frac{\omega}{(s+a)^2+\omega^2}$	damped sinusoid
$e^{-at}\cos(\omega t)\sigma(t)$	$\frac{s+a}{(s+a)^2+\omega^2}$	damped cosinusoid

Fig. 1 Mouse screaming for ice cream